

Swetha Ashok

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Education

University of California, Berkeley

Berkeley, CA

B.S. Electrical Engineering and Computer Science • GPA 3.6 • Expected Graduation May 2028

Relevant Coursework: Python, Java & Data Structures, Signals & Systems, Discrete Math and Probability Theory

Expected Completion Spring 2026: Circuits & Devices, PCB Design, C & Computer Architecture

Skills & Interests

Hardware: PCB Design, soldering, sensor testing, subsystem integration, debugging

Software: Python, Java, C, RISC-V, Github, MS Suite, LaTeX

CAD: SOLIDWORKS, Onshape, Autodesk Fusion 360, ANSYS (FEA), KiCAD

Languages: English (Native), Tamil (Native), Spanish (Conversational)

Interests: Audio Mixing, Violist and Audio Director of university film orchestra

Technical and Leadership Experience

Calsol – Solar Racing Team

Solar Project Manager (Current)

August 2024 – PRESENT

- Lead a 20+ member team responsible for solar module wiring, bypass diode soldering, and cell integration.
- Integrating the solar array into battery via MPPT, coordinating with electrical and mechanical teams on cross-functional requirements.
- Optimized rib and lightening holes layout using SOLIDWORKS and ANSYS structural simulations, reducing panel weight and enhancing structural integrity.

Society of Women Engineers

Academic Chair (Current) • Advocacy Chair

August 2024 – PRESENT

- Encouraged students to pursue higher education by coordinating events that connected them with professors, graduate students, and lab researchers.
- Advocated for underrepresented engineering students and served as the collegiate representative for the National SWE Disability Inclusion Affinity Group, organizing resource fairs and policy discussions that increased awareness and participation among members.
- Strengthened cross-organizational collaboration within the College of Engineering by planning and hosting semesterly social and professional events, which fostered networking and community building

FIRST Robotics Competition Team 2643

President • Project Manager • Safety Captain

August 2020 – May 2024

- Directed the design, wiring, and testing of multi-sensor robotic systems using SOLIDWORKS, enabling the 50-member team to build functional prototypes and competition ready robots that met various specifications.
- Maintained the robot's master CAD assembly and coordinated mechanical, electrical, and software teams, ensuring design consistency and efficient build timelines.

Computer Science Mentors

CS 61A Content Mentor

August 2020 – May 2024

- Created worksheets with LaTeX to support students in the Structure and Interpretation of Computer Programs (Python) course at UC Berkeley, providing clear practice material that improved student understanding and boosted overall course satisfaction.

Projects

Retro Synth

Co-Designer and Developer

January 2026 – PRESENT

- Collaborated with a three-person team to design and build a fully function 12 note PCB synth.
- Designing schematics using KiCAD and managing BOM to meet performance and budget constraints.